

5.2 ADDITION OF UNITS OF TIME

Add units of time in hours

Example: Add 5 hours and 2 hours.

Solution:

$$\begin{array}{r} 5 \text{ hours} \\ + 2 \text{ hours} \\ \hline 7 \text{ hours} \\ \hline \end{array}$$

EXERCISE 49

(A) Solve the following:

(1) $\begin{array}{r} 4 \text{ hours} \\ + 3 \text{ hours} \\ \hline \end{array}$

(2) $\begin{array}{r} 12 \text{ hours} \\ + 17 \text{ hours} \\ \hline \end{array}$

(3) $\begin{array}{r} 25 \text{ hours} \\ + 16 \text{ hours} \\ \hline \end{array}$

(4) $\begin{array}{r} 32 \text{ hours} \\ + 18 \text{ hours} \\ \hline \end{array}$

(5) $\begin{array}{r} 36 \text{ hours} \\ + 25 \text{ hours} \\ \hline \end{array}$

(6) $\begin{array}{r} 29 \text{ hours} \\ + 38 \text{ hours} \\ \hline \end{array}$

(B) Add:

- (1) 32 hours and 22 hours. (2) 14 hours and 12 hours.
(3) 108 hours and 117 hours. (4) 125 hours and 135 hours.
(5) 385 hours and 323 hours. (6) 480 hours and 342 hours.

Solve real life problems involving units of time for addition in hours

Example 1: In January, Nadeem played 26 hours hockey and 13 hours cricket. What is the total time he spent in play for the month of January?

Solution:	Played hockey	26 hours
	Played cricket	+ 13 hours
	Total time spent	39 hours

EXERCISE 50

- (1) In a month Bilal spent for 13 hours at his grand mother's home. He spent 4 hours at aunt's place. How long did he spend at the two homes?
- (2) A farmer took 53 hours to reap wheat crop and 68 hours to reap rice crop in a week. How much time he spent altogether?
- (3) Alesha's mother cooked lunch in 3 hours and dinner in 4 hours for guests. What is the total time she spent in cooking?
- (4) In rainy season, on Wednesday rain for 2 hours, on Thursday for 4 hours and on Friday for 3 hours. What is the total time of rain in three days.

5.3 SUBTRACTION OF UNITS OF TIME

Subtraction of units of time in hours

Example: Subtract 8 hours from 10 hours.

Solution:

$$\begin{array}{r} 10 \text{ hours} \\ - 8 \text{ hours} \\ \hline 2 \text{ hours} \end{array}$$

EXERCISE 51

(A) Solve the following:

(1)
$$\begin{array}{r} 5 \text{ hours} \\ - 4 \text{ hours} \\ \hline \end{array}$$

(2)
$$\begin{array}{r} 12 \text{ hours} \\ - 7 \text{ hours} \\ \hline \end{array}$$

(3)
$$\begin{array}{r} 22 \text{ hours} \\ - 18 \text{ hours} \\ \hline \end{array}$$

(4)
$$\begin{array}{r} 37 \text{ hours} \\ - 22 \text{ hours} \\ \hline \end{array}$$

(5)
$$\begin{array}{r} 33 \text{ hours} \\ - 14 \text{ hours} \\ \hline \end{array}$$

(6)
$$\begin{array}{r} 49 \text{ hours} \\ - 39 \text{ hours} \\ \hline \end{array}$$

(B) Subtract:

(1) 6 hours from 12 hours. (2) 6 hours from 48 hours.

(3) 145 hours from 165 hours. (4) 195 hours from 300 hours.

(5) 486 hours from 723 hours. (6) 578 hours from 995 hours.

Solve real life problems involving subtraction of units of time in hours

Example: Saleem travelled from Karachi to Sukkur in 8 hours by road. Ahmed travelled from Karachi to Lahore in 18 hours. Who travelled more and how much?

Solution: Ahmed travelled **18 hours**

Saleem travelled **– 8 hours**

10 hours

So, Ahmed travelled 10 hours more than Saleem.

EXERCISE 52

- (1) A bus took 8 hours and a car took 6 hours to travel from Sukkur to Hyderabad. Which vehicle is faster and how much?
- (2) Majida takes 22 hours to make two dresses. If she makes one in 15 hours. How much more time she take to make the other dress?
- (3) Ahsan reads a story book in 5 hours and Azeem reads the same book in 9 hours. Who took more time to read the book and how much?
- (4) A T.V channel - A relay 14 hours game programme while another T.V channel - B relay 20 hours. Find who relay games more time and how much?
- (5) Saira's family left Hyderabad at 5:00 p.m and reached at Karachi at 7:30 p.m. How much time did they travelled?

6.1 GEOMETRICAL SHAPES

Recognize point, line segment and ray

Starter: On the board, the teacher will draw different points and label them A, B, C etc.

The teacher will then join two points in a straight line

And explain that a straight line between two points is called a line segment and can be written as

A B
Line segment AB or BA

$\overline{AB}$ or $\overline{BA}$

On the board the teacher will draw various points and join them up in straight lines and wavy lines. Students will be called randomly to identify which of these are line segments (emphasize that line segments are in straight lines between two points).

Ask some students to write the names of the identified line segments.

Point

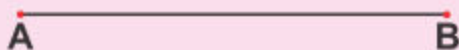
It used for representing of position of any place. Capital letters A, B, C, . . . , X, Y, Z are used to show the points. We can draw many points on a plain paper.

We can read them:

Point A, Point B, Point C, . . . Point Z.

**Line Segment**

A line segment is a straight path between two points. Here point A and point B are called end points.



We can call it either line segment AB or line segment BA. symbolically, we can write it as:

$\overline{AB}$ or $\overline{BA}$

$\overline{AB}$ is read as line segment AB and BA is read as line segment BA.

On the board the teacher will draw line segment

$\overline{AB}$

Then the teacher will extend one end point and draw an arrow mark to make the figure

$\overrightarrow{AB}$

Explain that this is called a ray. This has one end point, which is given in the example.

And is written as

$\overrightarrow{AB}$

$\overrightarrow{BA}$ or in this case B will be the end point

Call some children randomly to board and ask them to make line segments and label them.

Then call other students to the board and ask them to make rays from the line segments drawn on the board. Ask them to identify the end points.

Ray

Take a line segment $\overline{AB}$.

$\overline{AB}$

Now extend one end point of it, say point B in forward direction.

Draw an arrow mark. We get the figure

$\overrightarrow{AB}$

Teacher's Note

Teacher should give the concept of point, line segment and ray by drawings.

It is called ray AB. In symbol, it is written as $\overrightarrow{AB}$.

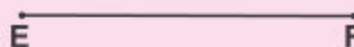
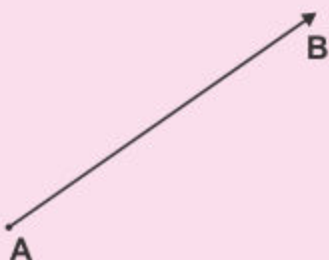
A ray has one end point. In $\overrightarrow{AB}$ has A is end point. Similarly we can draw $\overrightarrow{BA}$, B is end point.



A ray is formed by drawing an arrow mark on one end point of a line segment. $\overrightarrow{PQ}$ Ray PQ $\overrightarrow{QP}$ A ray has only one end point. The length of ray cannot be measured.

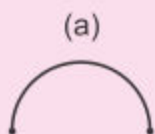
**Activity**

Tick (✓) the line segments and cross (✗) the rays. Also encircle (○) the points in each.



EXERCISE 53

- (1) Tick (✓) the line segment and cross the ray.



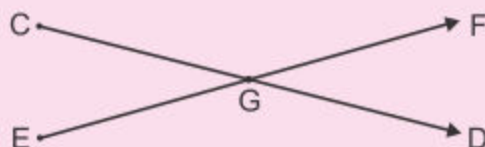
- (2) Write the end points of these figures.



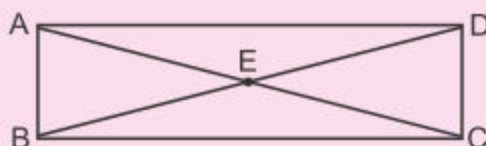
- (3) Write the name of this line segment in two ways:



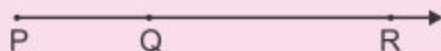
- (4) $\overrightarrow{CD}$ cross $\overrightarrow{EF}$ at what point?



- (5) Write the names of four line segments that have point A as common end point.



- (6) Write the names of all the rays and line segments shown in the figure.



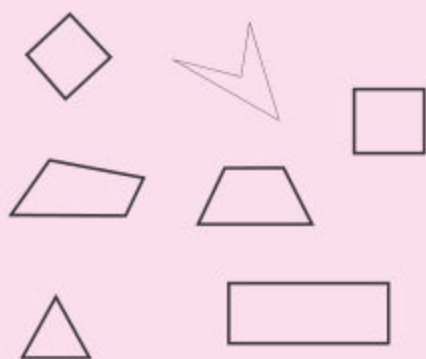
Classify figures according to number of sides as quadrilaterals (rectangles, squares) and triangles

Starter: On the board, the teacher will draw a rectangle and a square. Explain that any closed figure with four sides is called a quadrilateral.

Show the children that squares have four sides which are equal, but rectangles have opposite sides which are equal.

Choose a few children to come to the board and draw rectangles or squares.

The teacher will draw a variety of shapes and ask the children to identify a square, a rectangle and any other quadrilaterals.



Eventually the only shape left will be a triangle.

The teacher will explain that a triangle is a closed figure with three sides.

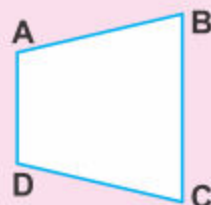
Quadrilateral

A quadrilateral is a closed figure. It has four sides and four vertices.

ABCD is a quadrilateral. Its sides are $\overline{AB}$, $\overline{BC}$, $\overline{CD}$ and $\overline{DA}$.

Points A, B, C and D are vertices.

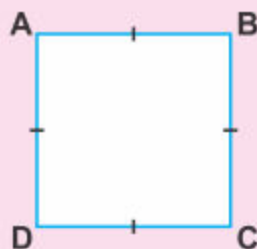
Squares and rectangles are quadrilateral.

**Square**

A quadrilateral whose four sides are equal in measure is called a square. ABCD is square.

$$\overline{AB} = \overline{BC} = \overline{CD} = \overline{DA}$$

A square has four vertices ABC and D.

**Rectangle**

A quadrilateral whose opposite sides are equal is called a Rectangle. PQRS is a

rectangle in which $m\overline{PQ} = m\overline{RS}$ and $m\overline{PS} = m\overline{QR}$

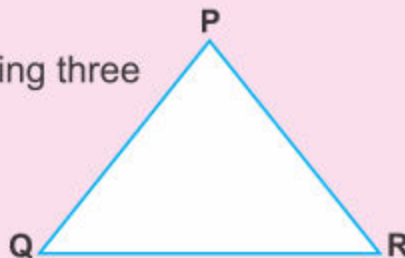
A rectangle has four vertices P, Q, R and S.

**Triangles**

A triangle PQR is a closed figure having three sides

$\overline{PQ}$, $\overline{QR}$ and $\overline{RP}$.

The points P, Q and R are called the vertices of the triangle PQR.





Activity Name the following quadrilaterals by joining dots. also name these shapes.

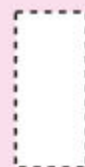
(a)



(b)



(c)



(d)



(e)



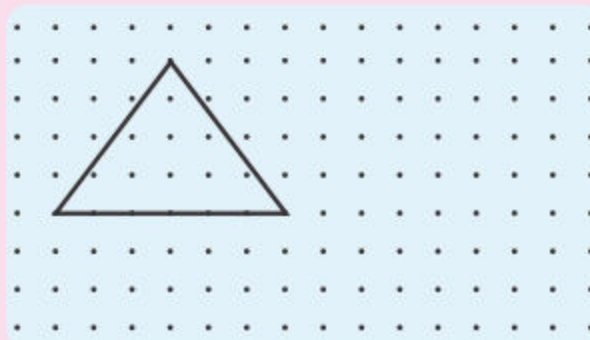
(f)



Example: Draw the triangle with the help of geo-board.



This is triangle, now we draw triangle on the geo-board bellow

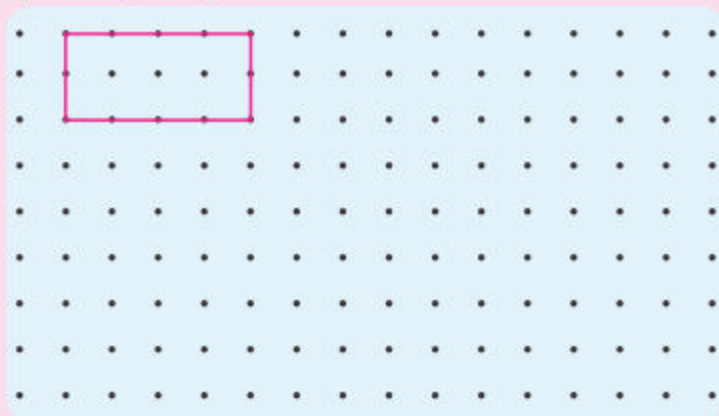


EXERCISE 54

- (1) Name the following quadrilaterals by joining dots. also name these shapes.



- (2) Draw two rectangles, two squares and two triangles by joining the given dots.



- (3) Complete the following sentences by Square, Rectangle, Triangle.

- (i) A plain figure bounded by 4 straight lines is called _____.
- (i) A closed figure bounded by 3 sides is _____.
- (iii) Opposite sides of a _____ are equal in measure.
- (iv) All sides of a _____ are equal in measure.
- (v) A quadrilateral whose opposite sides are equal is _____.